



Feasibility investigation of tuned mass damper for vibration control of curved floating bridge in winds and waves

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ABSTRACT

This paper investigates the feasibility and operability of a tuned mass damper (TMD) to control the lateral vibrations of a curved floating bridge. A curved floating bridge was designed with a bridge girder, 38 columns, and 19 pontoons, which was the same concept proposed for crossing Bjørnafjorden in Norway, except that the cable-stayed bridge was excluded in the present model. TMD was then attached to the mid-length of the bridge girder. Time-domain model was built to consider the girder-column-pontoon interaction based on the Cummins equation for pontoons and the lumped-mass line model for the bridge girder and columns, while TMD was modeled by the mass-spring-damper system. First, the genetic algorithm was employed to optimize the spring and damping coefficients of TMD at a fixed TMD mass under white noise wave excitations, and the fitness function was a summation of the standard deviation of the sway motion of 19 pontoons. Then, its feasibility was evaluated by comparing lateral vibrations and vertical bending moments with and without TMD under various combinations of wind/wave/current loads. Results demonstrate that TMD contributes to a reduction in the lateral vibration when resonance at the second wet natural frequency occurs, which can improve passenger comfort. Wind-induced lateral motions are significantly reduced at the second wet natural frequency while TMD has limited effects under other environmental loads. Moreover, bending moments are governed by higher modes mostly induced by storm waves, which cannot be controlled by TMD effectively.

1. Introduction

Floating bridges are one of the novel marine infrastructures, which is regarded as feasible alternatives to traditional cable-stayed or suspension bridges. In particular, when the waterway is long and deep, floating bridges have several critical advantages over conventional bridges in that the length of columns decreases significantly by introducing pontoons, which makes the entire bridge lighter, and installation can be easier by minimizing the foundation work (Dai et al., 2021). Floating bridges exist worldwide—Evergreen Point Floating Bridge in the USA, Nordhordland Bridge in Norway, Yumemai Bridge in Japan, William R. Bennett Bridge in Canada, Hobart Bridge in Australia, and Demerara Harbour Bridge in Guyana are representative examples (C. Jin et al., 2020). However, an optimal design that satisfies both safety and cost-effectiveness is hard to achieve, especially for long-span floating

bridges, for which large elastic responses induced by waves and winds are inevitable.

In recent years, active research has been conducted to understand and reduce hydroelastic and structural behaviors of different types of floating bridges. In particular, regarding wave loads, Cheng et al. (2018) (Z. Cheng et al., 2018) compared the dynamic and structural responses of a curved floating bridge suggested for Bjørnafjorden in Norway under inhomogeneous linear and nonlinear wave excitation and viscous drag loads, demonstrating that difference-frequency wave-excitation and viscous drag loads induce large lateral motions, while heave motions are mainly influenced by the linear wave excitation load. Cheng et al. (Z. Cheng et al., 2018) additionally investigated the effects of inhomogeneous wave loadings in detail to evaluate dynamic and structural responses of the same curved bridge; the results show that responses of the floating bridge are sensitive to wave condition/inhomogeneity. Viuff

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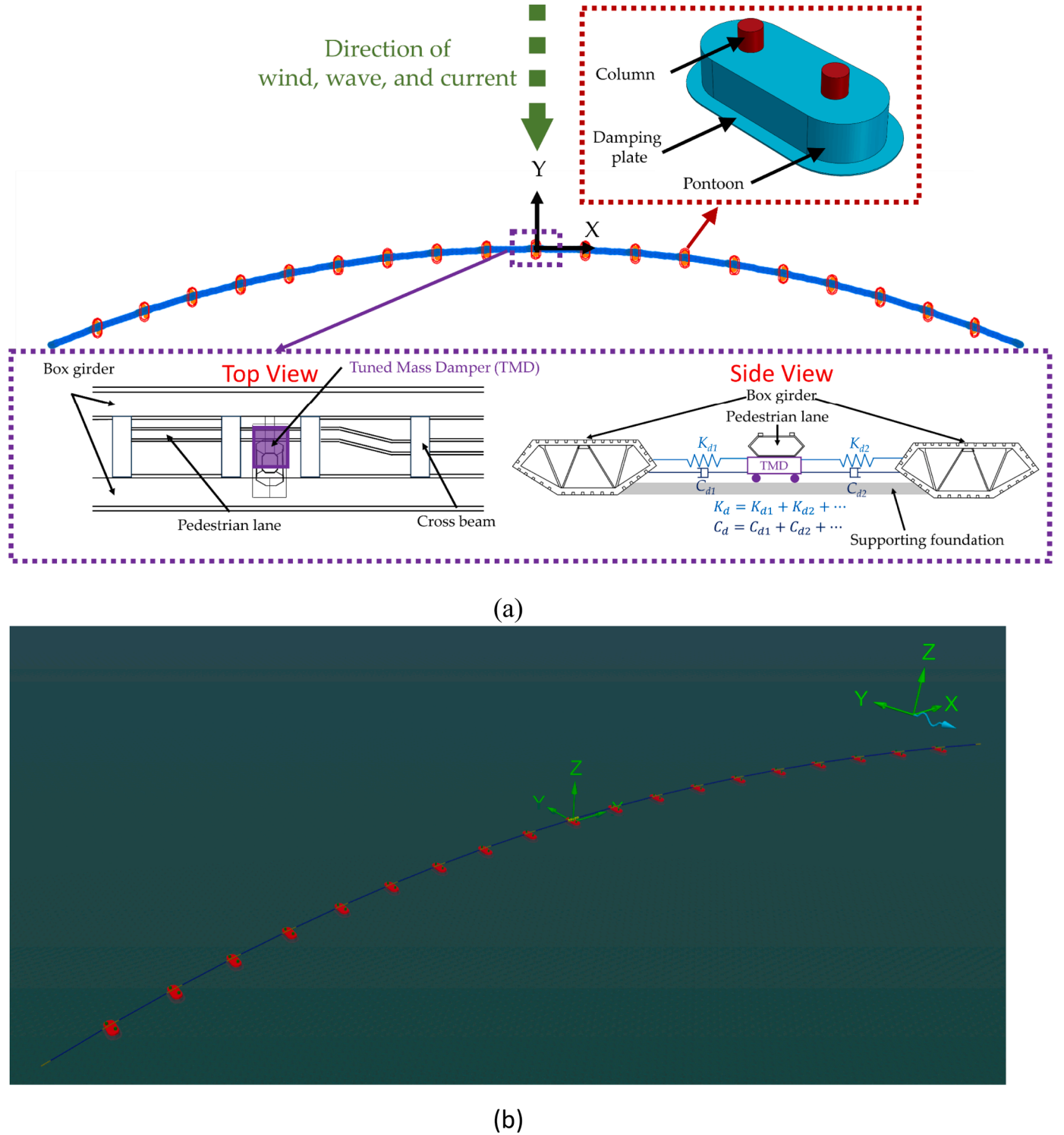


Fig. 1. General layout of a curved floating bridge with TMD: top and side views (a), 3D view (b).

et al. (Viuff et al., 2019) checked the influences of the wave direction on bridge responses for curved floating bridges. Dai et al. (Dai et al., 2021) evaluated fatigue damage of mooring lines for a straight-pontoon floating bridge for Bjørnafjorden under inhomogeneous wave loads. Kaligatla and Sahoo (Kaligatla and Sahoo, 2019) introduced bottom undulation to reduce wave-induced loads on a floating bridge, showing that sea-bed undulation plays a role in reducing the wave excitation load. Moreover, wind loads are also an additional element in the dynamics of long-span floating bridges since their first several lateral

natural frequencies are lower than the frequency range of typical wave spectra but closer to the dominant frequency ranges of wind energy. Jin et al. (C. Jin et al., 2020) considered several load combinations of linear/second-order wave-excitation, current, and wind loads to evaluate the dynamics of the curved floating bridge, claiming that wind loads induce larger lateral motion than second-order wave loads, whereas the linear wave-excitation load plays the most role in vertical responses. Cheng et al. (Cheng et al., 2019) conducted simulations under various combinations of steady wind, turbulent wind, wave, and

Table 1

Key parameters of the pontoon.

Item	Value	Unit
Width	28	m
Length	68	m
Height	14.5	m
Mass	11,300	t
Roll mass moment of inertia	4,900,000	t·m ²
Pitch mass moment of inertia	1,360,000	t·m ²
Yaw mass moment of inertia	5,700,000	t·m ²
Surge added mass at infinite frequency	6151	t
Sway added mass at infinite frequency	1913	t
Heave added mass at infinite frequency	38,980	t
Roll added mass at infinite frequency	11,670,000	t·m ²
Pitch added mass at infinite frequency	2,938,000	t·m ²
Yaw added mass at infinite frequency	1,271,000	t·m ²
Center of gravity	(0, 0, -4.2)	m
Center of buoyancy	(0, 0, -5.4)	m
Heave stiffness	17.5	MN/m
Roll water plane stiffness	5,700	MNm/rad
Pitch water plane stiffness	1,000	MNm/rad

Table 2

Key parameters of the girder and column.

Item	Value			Unit
	Girder S1	Girder F1	Column	
Mass	31,800	26,712	8,000	kg/m
EI_z	2.1×10^{14}	1.5×10^{14}	1.4×10^{12}	N·m ²
EI_y	3.7×10^{12}	2.6×10^{12}	1.4×10^{12}	N·m ²
EA	5.0×10^{11}	3.7×10^{11}	1.8×10^{11}	N
GI_x	3.5×10^{12}	2.8×10^{12}	1.1×10^{12}	N·m ²

* In Table 2, E and G are Young's and shear moduli, A is the cross-sectional area, I_y and I_z are the area moments of inertia about the weak and strong axes, respectively, while I_x is the area moment of inertia about the torsional axis.

current, supporting the importance of wind excitations on behaviors of the curved floating bridge. With similar curved bridge design as Cheng et al. (Cheng et al., 2019), Sha et al. (Sha et al., 2018) showed that wind excitations play more role in lateral motion than wave excitations. One of the common conclusions in these studies was that significant dynamic and structural behaviors are inevitables under wave and wind excitations associated with resonances, which leads to the necessity of coming up with strategies for mitigating floating bridge dynamics.

There exist several methods to handle the resonance of floating structures. Changes in the system's mass and/or stiffness can be one primary method (Yanik et al., 2016). For example, the material, size, or thickness of the girder or pontoon can be altered to place a target natural frequency outside the dominating frequency range of environmental loads. Finding an acceptable natural frequency is, however, challenging since not only low-frequency tsunamis and winds but also high-frequency waves and earthquakes co-exist in the ocean. The economic aspect is also an impediment to modifying natural frequency because making a very rigid system demands large capital and operational expenditures. An alternative method of vibration mitigation is to employ a vibration control system. While several passive, semi-active, and active vibration control methods have been proposed in coastal and ocean structures, including the application of heave damping plates, tuned mass dampers (TMDs), tuned liquid column dampers (TLCDs), and magnetorheological (MR) dampers (Wu et al., 2016; Lee et al., 2006; Wu et al., 2011; Zribi et al., 2004; Jin et al., 2021; Jang et al., 2019), many experts regard TMDs as one of the verified vibration-control technologies. TMDs have been extensively researched for vibration reduction of the ocean and coastal structures (Jin et al., 2021; Pourzangbar and Vaezi, 2022), as well as buildings (Kaveh et al., 2020). Refs. (Gutierrez Soto and Adeli, 2013; Kandasamy et al., 2016) provide extensive reviews of vibration control for the ocean and coastal structures and buildings.

Finding optimal TMD parameters is a critical step of the TMD design in that closed-form expressions (Den Hartog JP 1985; Warburton, 1982) are only available when there is no damping on the main structure and/or the system is linear. For highly non-linear systems like floating bridges, time-domain optimization is necessary (Jin et al., 2021; Bekdaş and Nigdeli, 2013). In addition, a variety of optimization methods for TMD parameters have been proposed, including the random decrement method, metaheuristic algorithms such as genetic algorithm (GA), particle swarm optimization (PSO), and harmony search (HS), and machine learning methods (Lin et al., 2001; Bekdaş and Nigdeli, 2011; Yucel et al., 2019; Jin et al., 2022). Furthermore, mode shapes are often needed to find optimal parameters for a multi-degrees-of-freedom (MDOF) system (Sadek et al., 1997).

This paper represents the feasibility of TMD for controlling the lateral vibration of a curved floating bridge under wind and wave excitations. While TMD applications to similar infrastructure such as conventional bridges (Tubino and Piccardo, 2015; L. Wang et al., 2021) or submerged floating tunnels (Jin et al., 2021; Jin et al., 2022) have been investigated, its application to a floating bridge was not found in the open literature. In addition, the one of challenges of the floating bridge compared to the existing conventional bridges is large dynamic responses under surrounding environments. Hence, this paper provides insights into potential future applications of TMDs for the floating bridge by ensuring their operability and feasibility. The present study considered a passive TMD since the objective is to understand the overall performance of TMD for floating bridges. Adaptive-passive, semi-active, or active TMDs can also be applicable to achieve improved performance, as proven under the wind, earthquake, and human-induced vibrations, in that accommodating the change in the system or environmental conditions is possible (Y. Wang et al., 2021; L. Wang et al., 2023; Wang et al., 2020; Wang et al., 2019; L. Wang et al., 2023; L. Wang et al., 2023). This paper consists of 6 Sections; in Section 2, the design of a curved floating bridge with TMD is introduced; in Section 3, the theory and formulation of the time-domain model are introduced; in Section 4, the optimization of TMD parameters under white-noise wave excitations is presented; Section 5 presents the TMD effects on vibration reduction of the floating bridge under different combinations of wind and wave excitations, while Section 6 gives concluding remarks and future work.

2. Design of curved floating bridge with tuned mass damper

Fig. 1 depicts the general layout of a curved floating bridge, and Tables 1-2 provide the important design parameters. The present curved floating bridge was similarly modeled to one proposed for Norway's Bjørnafjorden, and the geometry and properties of the original curved floating bridge are given in Ref. (Larsen, 2016). The present design was, however, slightly modified such that (1) a cable-stayed bridge at the left end was not taken into account and (2) an equal-column-height bridge was designed for simplicity, whereas the original design had different heights of columns for facilitating ship traffic. The present floating bridge consisted of a bridge girder, 19 pontoons, 38 columns, and abutments at both ends. The girder was made of steel and concrete and had a total length of 3940 m. A girder consisted of two box girders for car traffic, one pedestrian lane for people, and multiple cross beams. The equivalent axial, bending, and torsional stiffness were obtained from the complex geometry, as given in Table 2. There were two types of girders: S1 and F1. The S1 type, which had thicker plates than the F1 type, was used only at the connection locations with columns, while the remaining sections were modeled with the F1 type. Along the bridge girder, 19 pontoons were evenly distributed with an equal distance of 197 m between them. The length, height, and draft of the pontoon were 68 m, 14.5 m, and 10.5 m, respectively. A damping plate with a thickness of 0.5 m was located at the bottom to dampen non-planar (heave, roll, and pitch) motions. Each pontoon and the girder were connected by two circular hollow steel columns. The diameter and length of the columns were 8 m and 7.5 m, respectively. Two columns were 37 m apart and

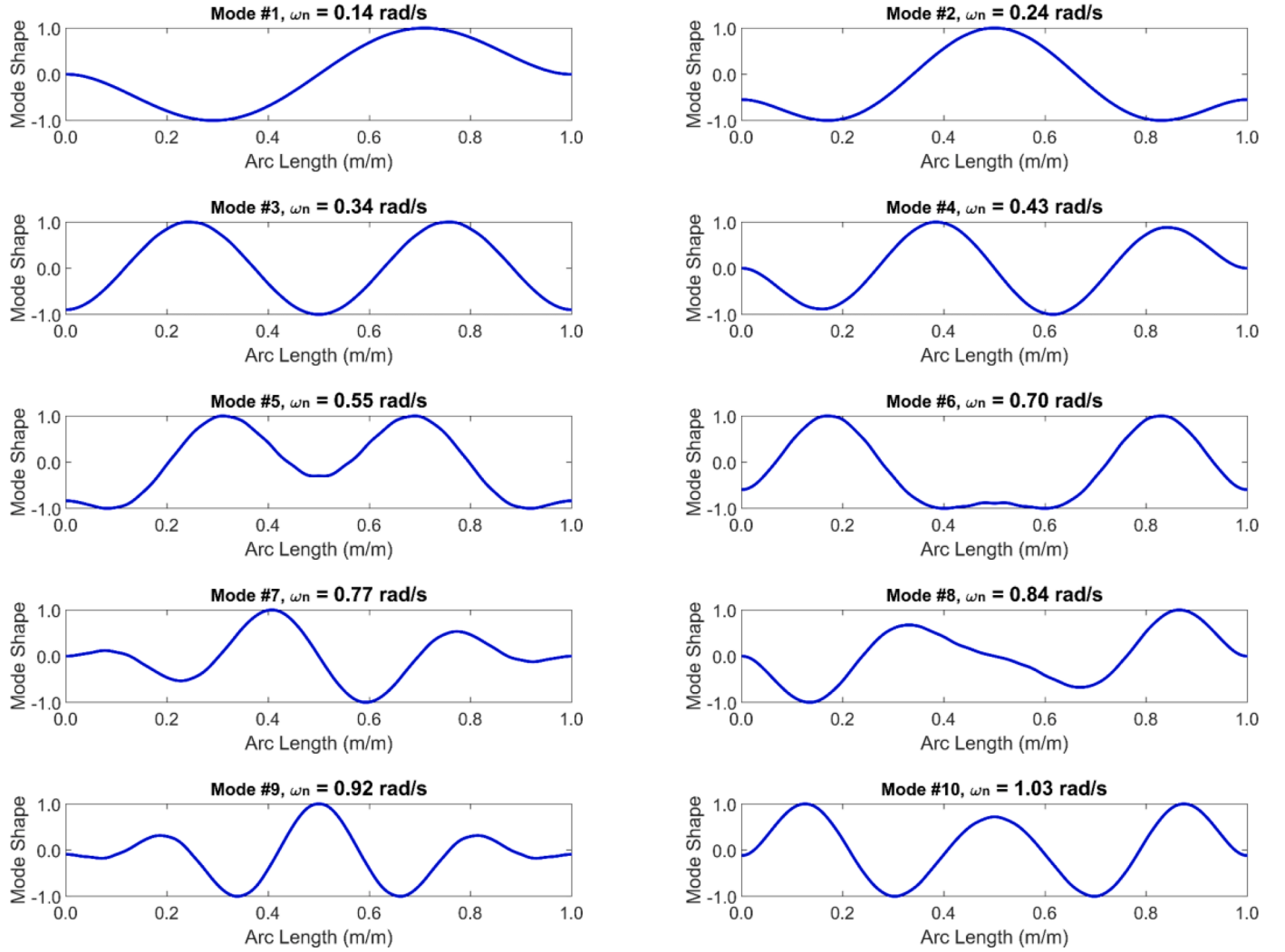


Fig. 2. Lowest 10 wet natural frequencies and corresponding mode shapes in the lateral direction without TMD.

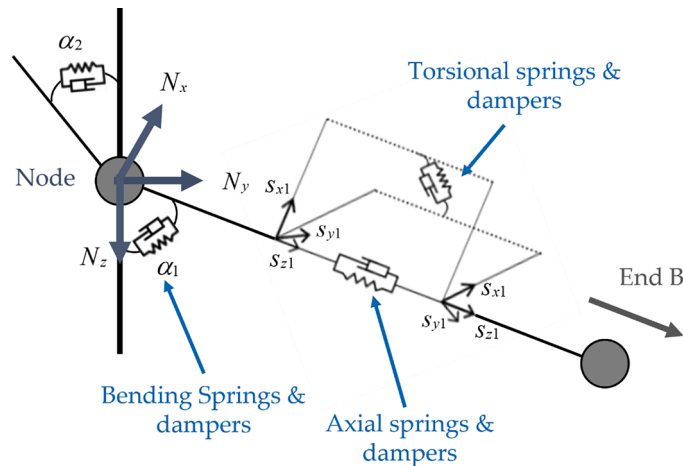
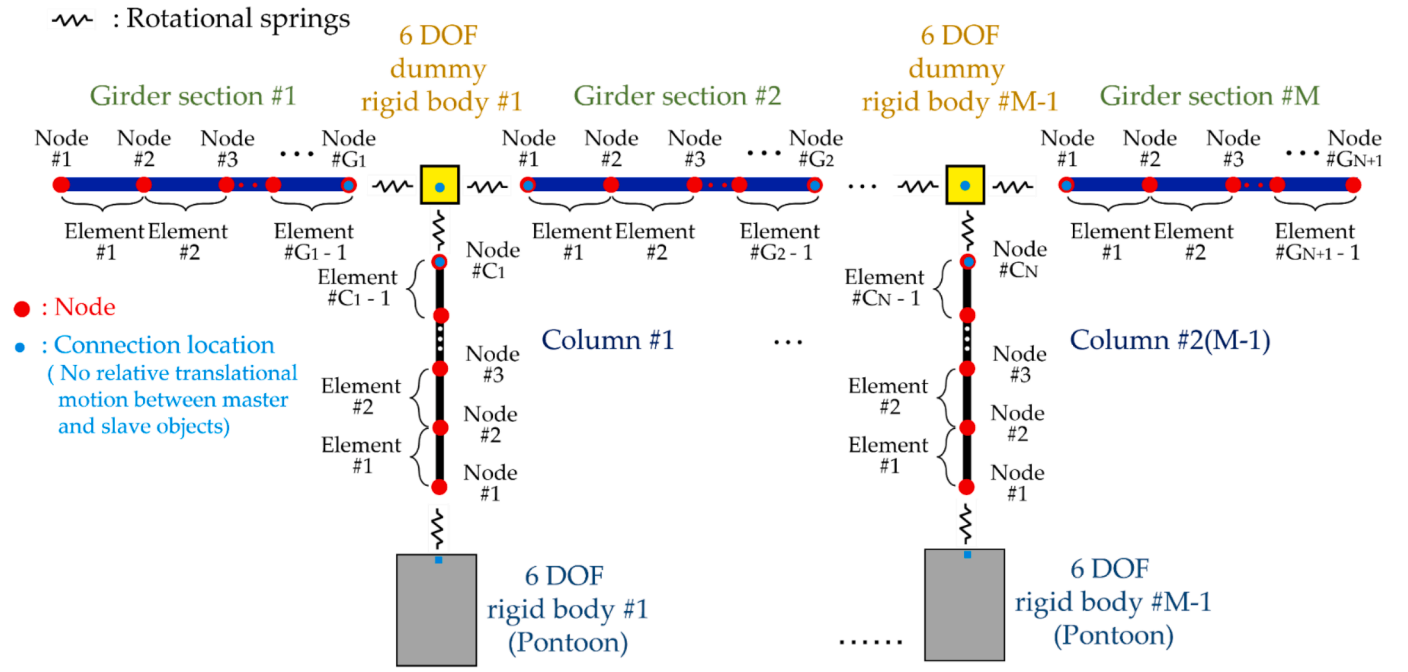


Fig. 3. Spring and damper arrangement for an element between two nodes (Orcina, 2018).

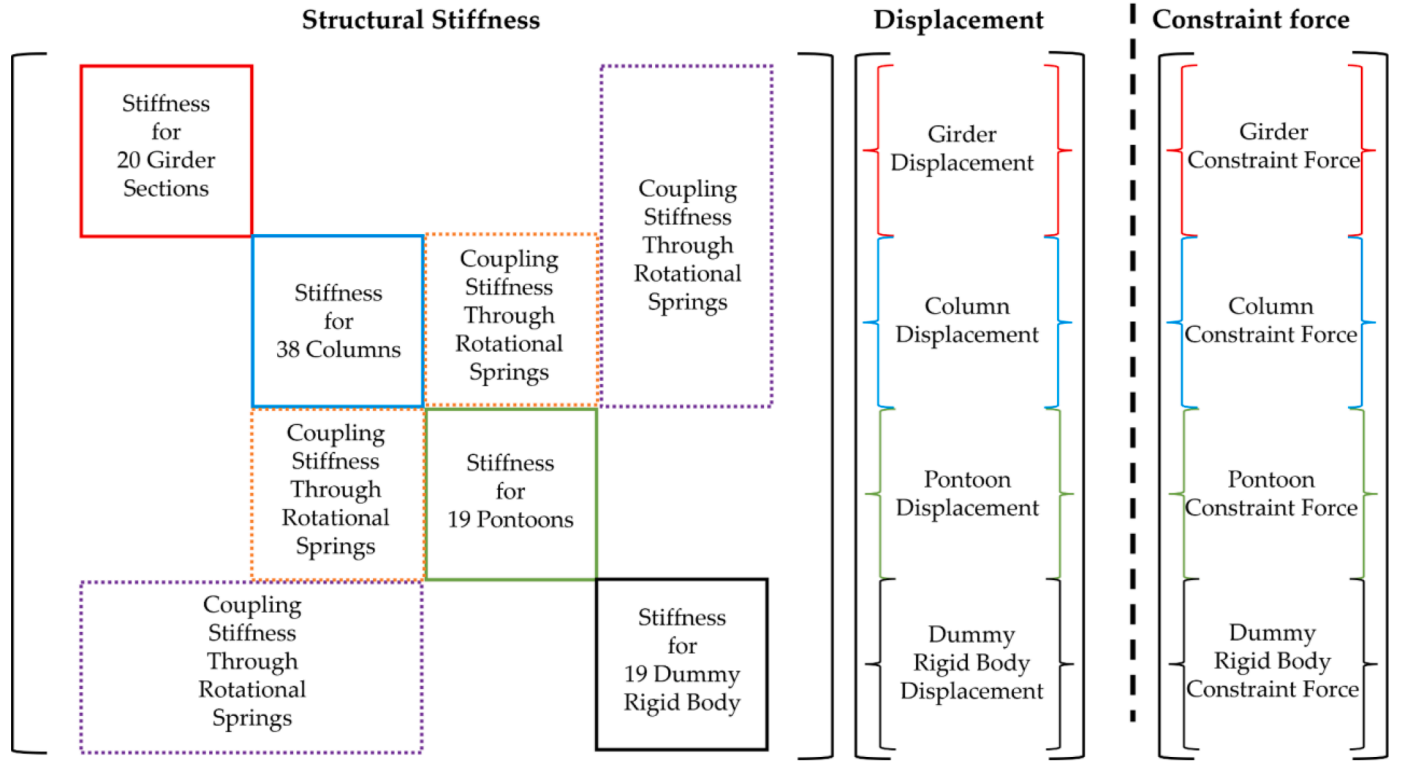
placed to be perpendicular to the axial direction of the girder. Since abutments were positioned at both ends, a fixed-fixed boundary condition was assumed at both ends. The lowest 10 wet natural frequencies of the floating bridge and corresponding mode shapes of the girder in the

lateral direction are presented in Fig. 2. In wet mode calculations, a respective natural frequency was calculated at an added mass matrix at that natural frequency, which required iterations since the added mass was frequency-dependent. Mode shape calculations were conducted after static simulation under full environmental conditions, including storm and ocean swell waves with significant wave heights of 3 m and 0.4 m, respectively, a mean wind velocity of 22.7 m/s, and a surface current velocity of 0.7 m/s. Even though the design is similar to one proposed for Norway's Bjørnafjorden, the elimination of the cable-stayed bridge and design modification increase the fundamental lateral natural frequency and change the mode shapes. In particular, the obtained natural frequencies are slightly higher than that in Ref. (Z. Cheng et al., 2018) due to the elimination of the cable-stayed bridge and reduced bridge length. Note that due to this reason, the optimized TMD parameters for that floating bridge will differ from those of the current bridge structure.

TMD consisted of mass, springs, and dampers. Since there was no actual application of TMD to floating bridges, it was assumed that TMD mass, made of steel, was 40 m long and positioned at the mid-length of the girder. The location of TMD and its connection with the main bridge structure are illustrated in Fig. 1(a). At the mid-length of the girder, there was empty space between two box girders. Thus, the supporting foundation was placed between two box girders, while TMD was positioned on top of the supporting foundation. TMD was then connected to the main structure through multiple springs and dampers. It was



(a)



(b)

Fig. 4. Dummy-connection-mass method for solving coupled interaction (a) and total stiffness matrix (b) without TMD (Jin and Kim, 2018).

designed to allow lateral motion only, while the other 5 DOF motions were synchronized with the bridge girder's translational and rotational motions. TMD was designed to be equipped with 10 springs and 10 dampers connected to the girder. TMD parameters need to be optimized, and the optimization process is detailed in Section 4.

3. Theory and formulation of time-domain dynamics model

3.1. Floating bridge

The time-domain dynamics simulation model to solve the girder-pontoon-column-TMD coupled interaction was constructed by the

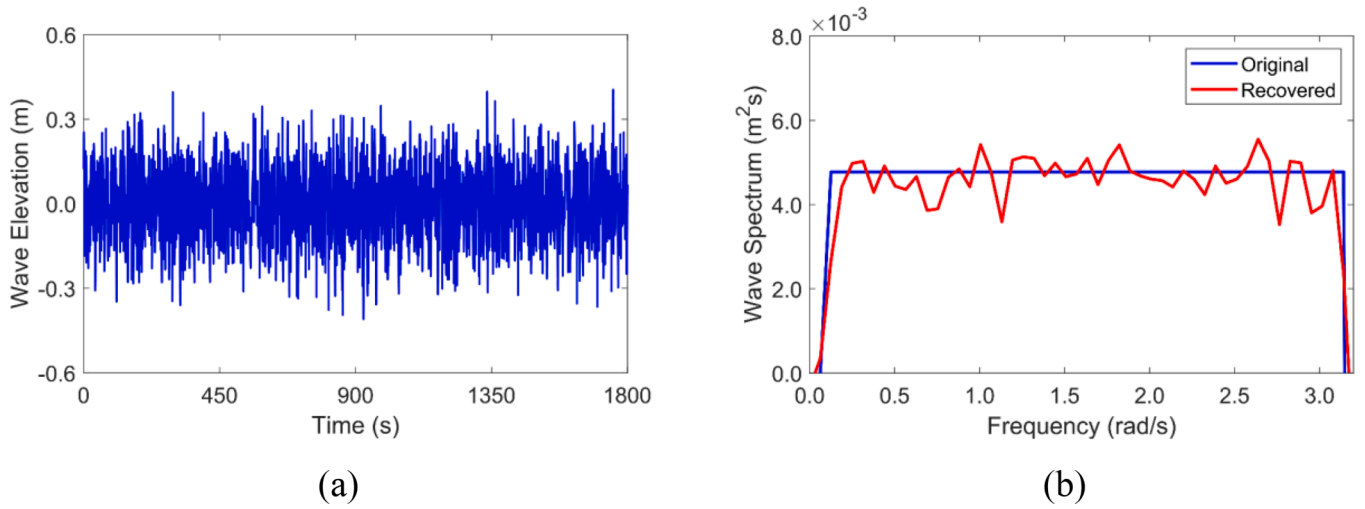


Fig. 5. Time history of white-noise wave elevation (a) and corresponding original spectrum and recovered spectrum from FFT (b).

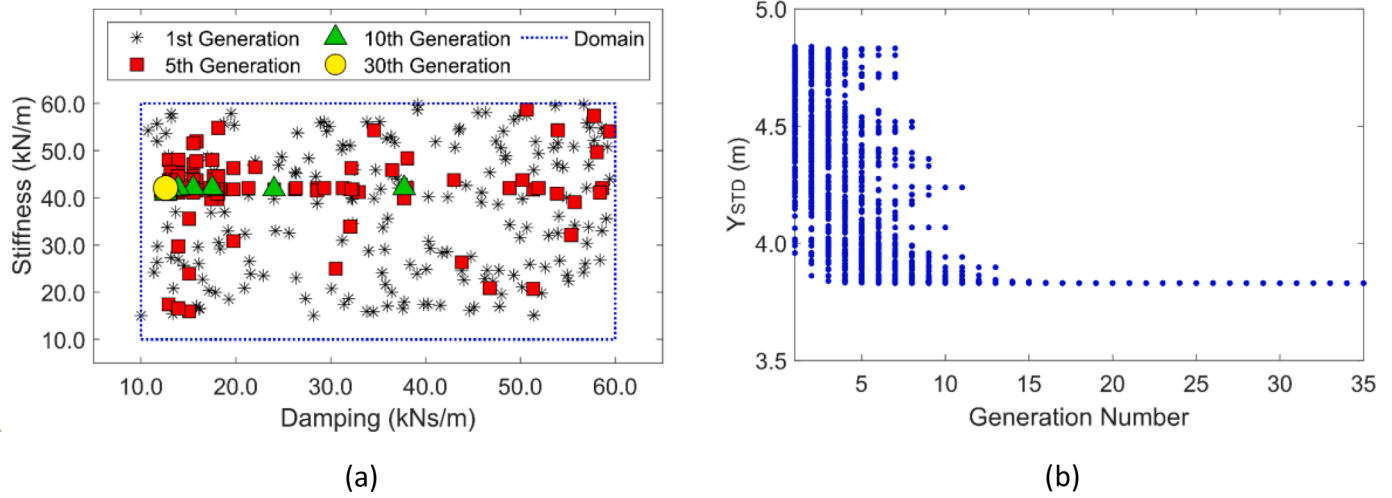


Fig. 6. Population distributions at the different numbers of generations (a) and distribution of fitness values (i.e., summation of the standard deviation of sway motion of 19 pontoons) as a function of the number of generations (b).

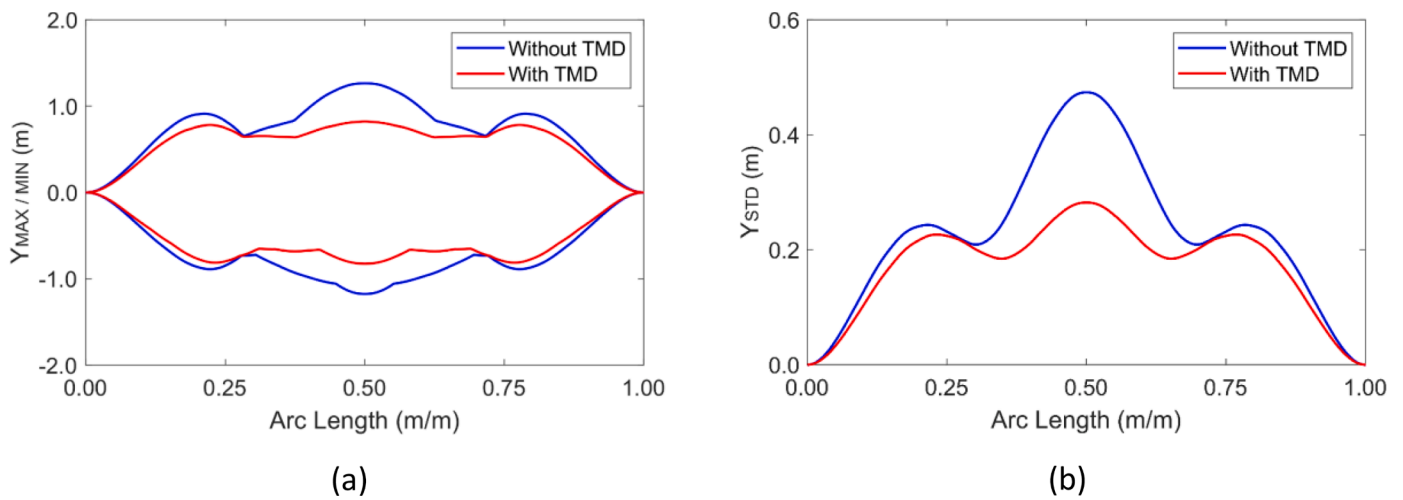


Fig. 7. Maximum, minimum (a), and standard deviation (b) of girder's lateral motion with and without TMD under white noise wave excitations.

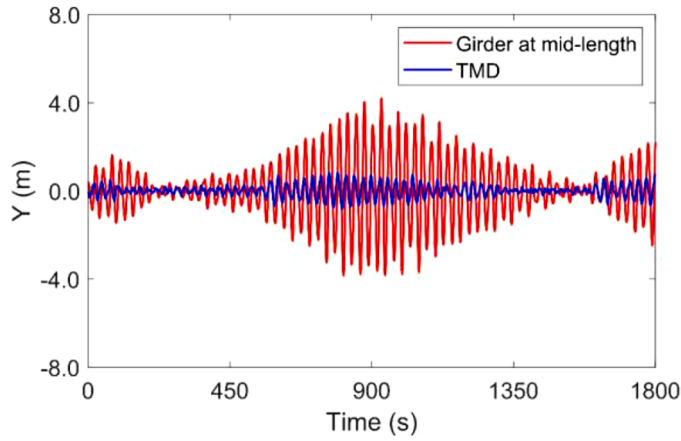


Fig. 8. Time histories of lateral motion of girder at mid-length and TMD under white noise wave excitations.

Table 3

Description of cases.

Case #	First-order wave	Second-order wave	Wind	Current
1	O	×	×	O
2	O	O	×	O
3	×	×	O	O
4	O	O	O	O

OrcaFlex program (Orcina, 2018). A matrix form of the equation of motion for this coupled dynamics system without TMD can be written as:

$$\mathbf{M}\ddot{\mathbf{X}} + \mathbf{C}\dot{\mathbf{X}} + \mathbf{K}\mathbf{X} = \mathbf{F}, \quad (1)$$

where $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ denote the matrices of the total mass, total damping, and total stiffness, respectively, and $\mathbf{F}$ and $\mathbf{X}$ represent the vectors of the total external loads and displacements with an upper dot representing the time derivative of a variable.

The 3D diffraction/radiation panel method was employed to obtain wave diffraction and radiation loads on the pontoons. That is, hydrodynamic coefficients and first- and second-order wave loads were obtained in frequency domain by solving the 3D boundary value problem with the Laplace equation as the governing equation and free-surface, body, far-field, and bottom boundary conditions. Cummins equation was, then, used as the time-domain governing equation of motion for pontoons as:

$$[\mathbf{M}_p + \mathbf{M}_p^a(\infty)]\ddot{\mathbf{X}}_p + \mathbf{K}_p^h\mathbf{X}_p = \mathbf{F}^1 + \mathbf{F}^2 + \mathbf{F}^c + \mathbf{F}^d + \mathbf{F}^i, \quad (2)$$

where a subscript p denotes a pontoon, $\mathbf{M}^a(\infty)$ is the added mass matrix at infinite frequency, $\mathbf{K}^h$ is the hydrostatic and gravitational restoring coefficient matrix, and $\mathbf{F}^1$, $\mathbf{F}^2$, $\mathbf{F}^c$, $\mathbf{F}^d$, and $\mathbf{F}^i$ are the vectors of the first-order wave excitation load, second-order difference-frequency wave load, the radiation damping load expressed by the convolution integral, viscous drag load induced by wind and current, and the interaction load with columns at their connection locations. $\mathbf{F}^2$ was obtained by Newman's approximation by which off-diagonal terms of the quadratic transfer function (QTF) were assessed by diagonal terms; the process of solving the second-order boundary value problem was eliminated since the first-order potential was only necessary. $\mathbf{F}^d$ was computed by the drag term of the Morison equation. Each body has 6DOFs. In addition, when there is a significant interaction between two bodies, a multi-body diffraction/radiation problem needs to be solved (C. Jin et al., 2020). However, since the distance between pontoons was large enough and the wave direction was assumed to be perpendicular to the pontoon's

lateral direction, where minimum interaction between pontoons was expected, a single pontoon was only considered to estimate diffraction and radiation loads. Since the formulations to calculate these components are well-known, it is not elaborated on here. Ref. (C. Jin et al., 2020) explained these formulations for the same curved floating bridge in detail.

The girder and columns were modeled by the lumped-mass-based line model in which each girder or column consists of a certain number of nodes and finite elements. While nodes consider all physical properties, axial, bending, and torsional behaviors are represented by linear axial, bending, and torsional springs between nodes, as shown in Fig. 3. These spring elements were considered in part of $\mathbf{K}$. In the line model, two bending springs and dampers are located on both sides of a node, whereas axial and torsional springs and dampers are positioned between two neighboring nodes. Based on linear springs, the following equations can be introduced to describe the effective tension and the magnitude of bending and torsional moments:

$$T_E = EA\varepsilon + (1 - 2\nu)(p_O a_O - p_I a_I), \quad (3)$$

$$|m_{BM}| = EI|\kappa|, \quad (4)$$

$$|m_T| = GJ\frac{\tau}{l_0}, \quad (5)$$

where EA , EI , and GJ denote the axial, bending, and torsional stiffness, ε denotes the total mean axial strain, κ represents the curvature, τ is the torsional angle, ν is the Poisson's ratio, l_0 denotes the unstretched element length, and p and a are pressure and cross-sectional area, respectively, with subscripts, I and O , representing the internal and external terms. Dynamic wind force acts on these components (i.e., girder and columns), which was considered by the Morison drag term. In addition, the traffic load in the vertical direction was also taken into account as a static load.

After each component was separately built, the coupled floating bridge model was finalized using a specially devised connection method called the dummy-connection-mass method, as shown in Fig. 4(a). Firstly, 6DOF dummy rigid bodies were placed wherever pontoons were located. The dummy rigid bodies did not contain any physical quantities since they were only used for connection with a negligible mass of 1 kg. Then, a bridge girder was decomposed into M girder sections since there were $M-1$ pontoons. Next, all end nodes of the neighboring girder sections and columns were rigidly connected to the corresponding dummy rigid body, by adjusting the boundary conditions of slave objects and introducing rotational springs. Translational motions of end nodes of lines (slave objects) rigorously followed those of the dummy rigid body (master object). This can be achieved by constraining translational motions of slave objects to be identical to those of the master object. Rotational motions of end nodes followed the dummy rigid body by adopting very stiff rotational springs. By having rotational springs in three axes and using very large values of rotational spring coefficients, no relative rotational motions occurred between the master object and slave ones, effectively establishing rigid connections. $\mathbf{F}^i$ occurs due to this interaction on pontoons, while an equal and opposite direction force acts on the counter bodies. Fig. 4(b) explains the connection between components in a matrix form. While each component has its own stiffness matrix, the coupling stiffness matrices are placed to simulate rotation springs for connections. Additionally, constraint forces $\mathbf{F}^i$ also appear owing to the coupling of translational motions.

3.2. Tuned mass damper

Next, TMD was modeled by a mass-spring-damper system. While TMD had 6DOFs, the lateral motion was only allowed to control lateral motion. Then, a matrix form of the equation of motion for the coupled system given in Eq. (1) can be extended with TMD DOFs as:

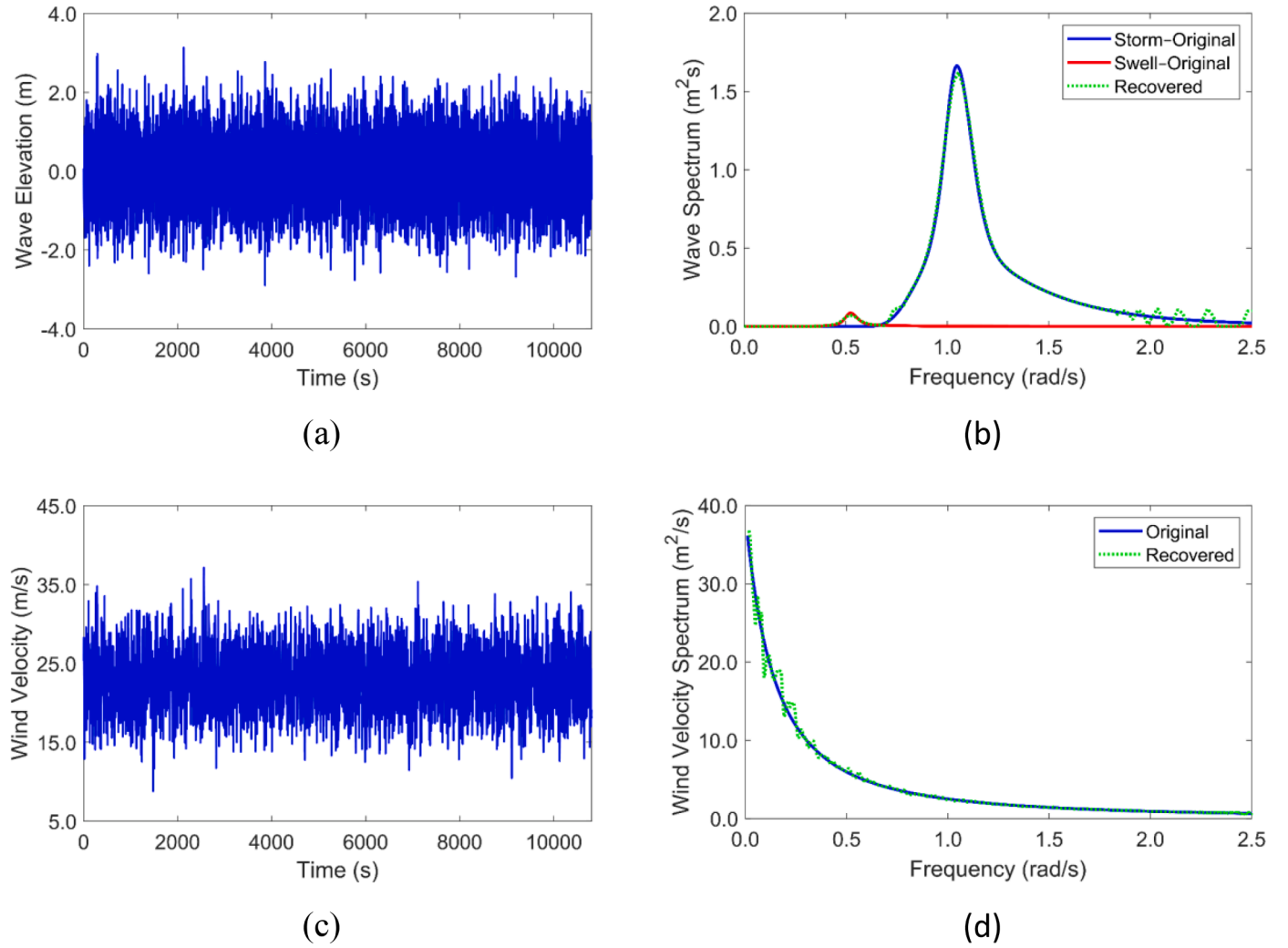


Fig. 9. Time histories of generated wave elevation (a) and wind velocity (c) and corresponding original spectra and recovered spectra from FFT (b, d).

$$\begin{bmatrix} \mathbf{M} & 0 \\ 0 & \mathbf{M}_d \end{bmatrix} \begin{Bmatrix} \ddot{\mathbf{X}} \\ \ddot{\mathbf{X}}_d \end{Bmatrix} + \begin{bmatrix} \mathbf{C} + \mathbf{C}_d & -\mathbf{C}_d \\ -\mathbf{C}_d & \mathbf{C}_d \end{bmatrix} \begin{Bmatrix} \dot{\mathbf{X}} \\ \dot{\mathbf{X}}_d \end{Bmatrix} + \begin{bmatrix} \mathbf{K} + \mathbf{K}_d & -\mathbf{K}_d \\ -\mathbf{K}_d & \mathbf{K}_d \end{bmatrix} \begin{Bmatrix} \mathbf{X} \\ \mathbf{X}_d \end{Bmatrix} = \begin{Bmatrix} \mathbf{F} \\ 0 \end{Bmatrix}, \quad (6)$$

where a subscript *d* denotes a component related to TMD, and $\mathbf{C}_d$ and $\mathbf{K}_d$ are the 6×6 matrices of TMD damping and stiffness. The lateral component (2,2) of $\mathbf{C}_d$ and $\mathbf{K}_d$ needs to be optimized for best-performed TMD. Other 5 DOF motions were synchronized with the girder's translational and rotational motions at the connection location through rigid connection, i.e., except for the (2,2) component, other diagonal terms of $\mathbf{K}_d$ were set to be $K_d(1,1) = K_d(3,3) = 5 \times 10^{15}$ N/m and $K_d(4,4) = K_d(5,5) = K_d(6,6) = 5 \times 10^{16}$ Nm not to allow relative motions between the floating bridge and TMD. In addition, in Eq. (6), the effect of TMD's gravitational force under roll motion, i.e., $\mathbf{M}_d g \sin(\theta)$ where θ is the girder's inclination angle, is not taken into account by assuming that its effect is small under small torsional motion. Then, the coupling between them is only induced by $\mathbf{C}_d$ and $\mathbf{K}_d$.

4. Optimization process of tuned mass damper parameters

The spring and damping coefficients of TMD were optimized to maximize the effectiveness of TMD. Optimization is inevitable in that there is no closed-form expression of TMD parameters when the system

is nonlinear and there exists damping in the main structure, such as the present floating bridge. This study adopted GA (Goldenberg, 1989) regarded as one of the most widely used metaheuristic algorithms. Many optimization problems in engineering were solved by GA (Guerrero et al., 2018; Gupta et al., 2018), including TMD optimization (Jin et al., 2021). Optimization by GA is processed by four major steps. (1) GA begins with a random generation of an initial population of individuals, called the first generation; GA evaluates the performance of initial populations through the fitness function (i.e., objective function); and the evaluated individuals are ranked. (2) Selection, crossover, and mutation processes are initiated to generate the new generation; the best-formed individuals are selected, and offspring is generated through additional crossover and mutations. (3) This process is repeated until satisfactory convergence is achieved. (4) Finally, the highest-ranked individual is selected as the optimum parameter. MATLAB GA function was adopted, considering the availability of coupling GA with the dynamics simulation model. Parameter optimization was achieved by coupling Matlab GA with the OrcaFlex model. The fitness function was selected to be the summation of the standard deviation of the sway motion (lateral motion) of 19 pontoons, which can be expressed as:

$$\text{fitness function} = \sum_{p=1}^{19} STD_{y,p} \quad (7)$$

where *p* denotes pontoon # (i.e., 19 pontoons), $STD_{y,p}$ is the standard deviation of lateral motion of *p*-th pontoon. The problem minimized the

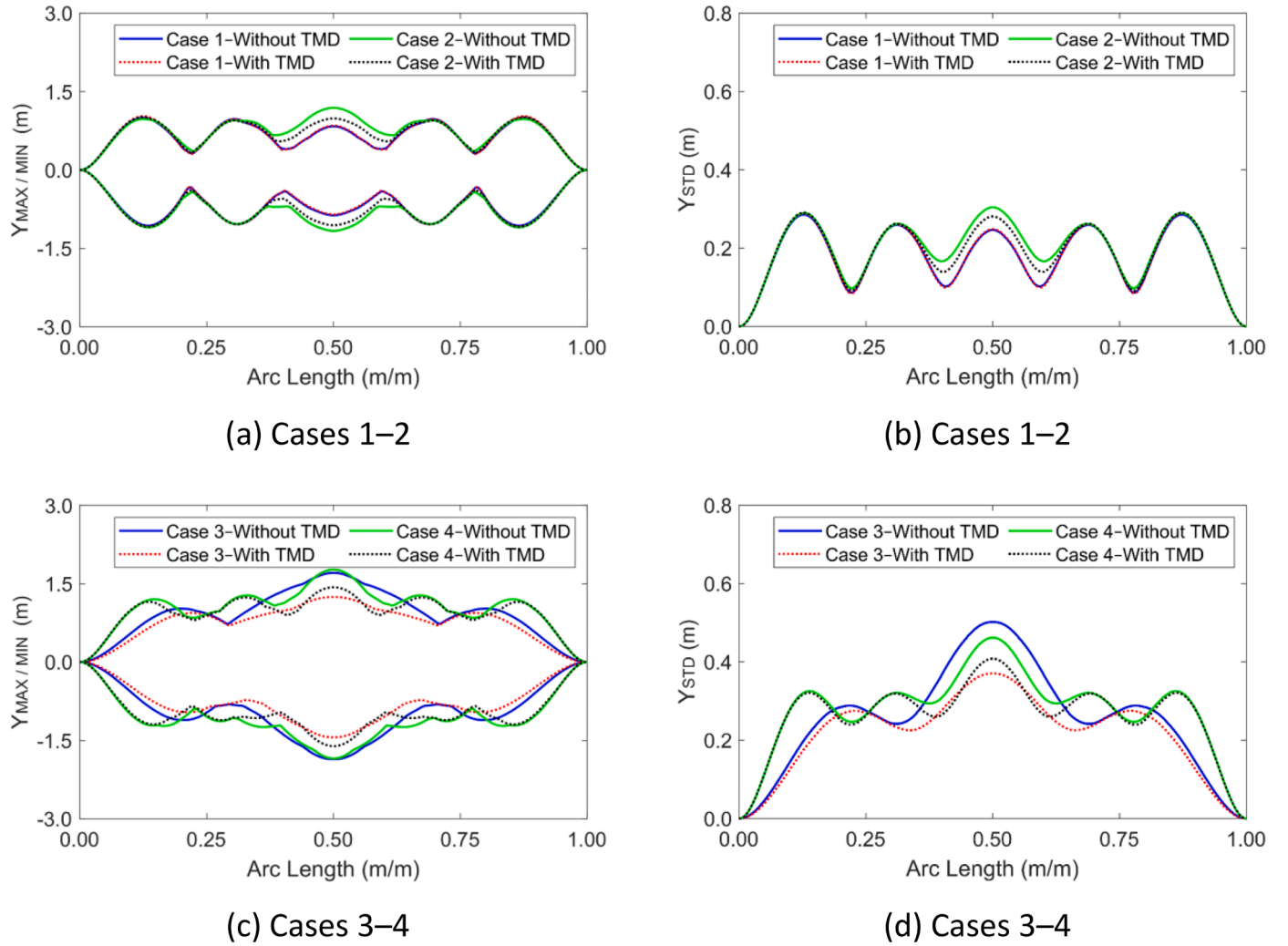


Fig. 10. Envelopes of maxima and minima (a and c) and standard deviations (b and d) of girder's lateral motions with and without TMD in Cases 1–4.

Table 4

Statistical values of lateral motion at the girder's mid-length.

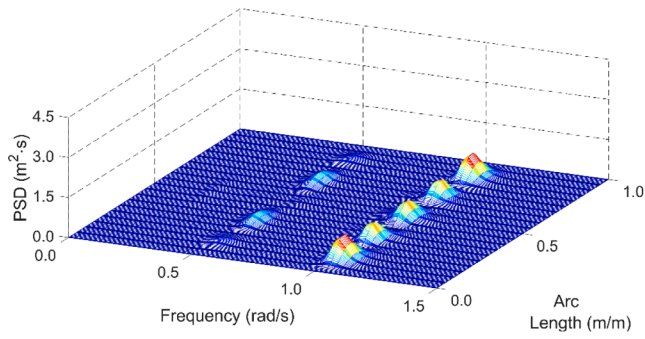
Case #	Maximum (m)			Minimum (m)			Standard deviation (m)		
	With TMD	Without TMD	Difference (%)	With TMD	Without TMD	Difference (%)	With TMD	Without TMD	Difference (%)
Case 1	0.847	0.834	1.5	−0.851	−0.869	2.0	0.248	0.246	0.8
Case 2	0.985	1.188	18.7	−1.061	−1.168	9.7	0.281	0.304	8.0
Case 3	1.246	1.706	31.2	−1.442	−1.860	25.3	0.371	0.502	30.1
Case 4	1.429	1.768	21.2	−1.612	−1.848	13.7	0.408	0.462	12.4

fitness function. While TMD mass was fixed at 1% of the modal mass, which is 692.64 tons, spring and damping coefficients were optimized through GA. Modal mass was based on the second wet mode, which was regarded as a major mode shape that intrigues strong resonant motions observed in previous investigations (C. Jin et al., 2020; Z. Cheng et al., 2018). The given TMD mass is on the same order of magnitude as that used for large-size TMDs applied to commercial buildings (Gutierrez Soto and Adeli, 2013). The biggest challenge of time-domain optimization is simulation time, so the population size was fixed to 208, while default values of MATLAB GA were chosen for other parameters. The search domains for spring and damping coefficients were [15, 60] kN/m and [10, 60] kNs/m, which were constraints in the optimization problem. These ranges are sufficient considering the dominant lateral natural frequency of a floating bridge of 0.24 rad/s under wave and wind excitations, as shown in the previous study (C. Jin et al., 2020). Wave loads

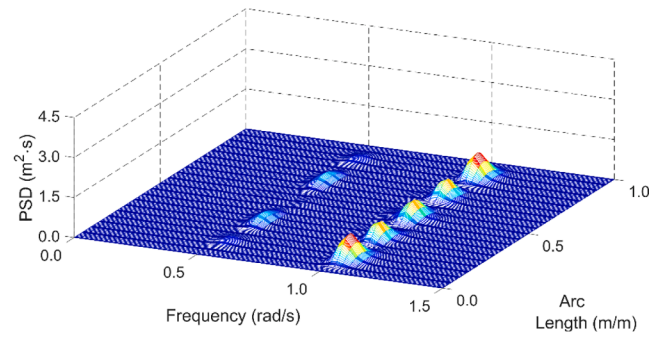
from the white-noise wave spectrum were inputted into pontoons in this optimization process. 30-minute simulations were conducted for time-efficient optimization. The lowest and highest cut-off frequencies were 0.13 rad/s and 3.14 rad/s. The time history of random waves from the white-noise wave spectrum was generated by superposing regular wave components. White-noise-based loading signals have been widely used in TMD optimization (e.g., (Chen and Tan, 2021; Marano et al., 2010; Rüdinger, 2006)). The following equation can describe the time history of random waves from the spectrum:

$$\zeta(y, t) = \sum_{j=1}^N \sqrt{2S(\omega_j) \Delta \omega_j} \cos(k_j y - \omega_j t + \phi_j) \quad (8)$$

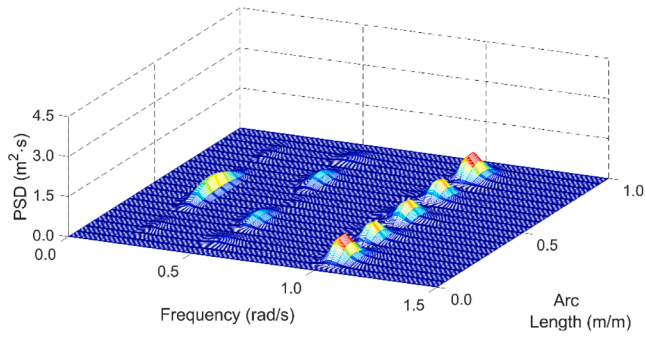
where N is the number of wave components, S is the wave spectrum, k and ω are wave number and angular frequency, ϕ is random phase



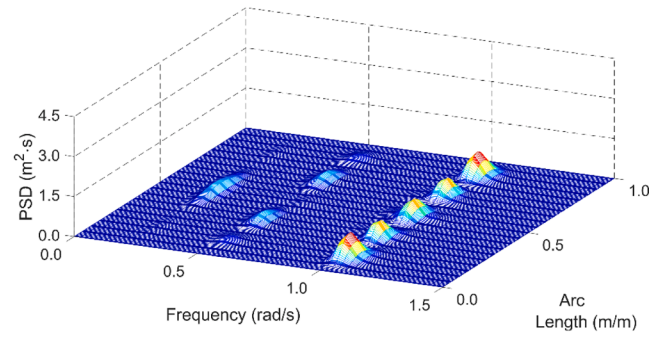
(a) Case 1 without TMD



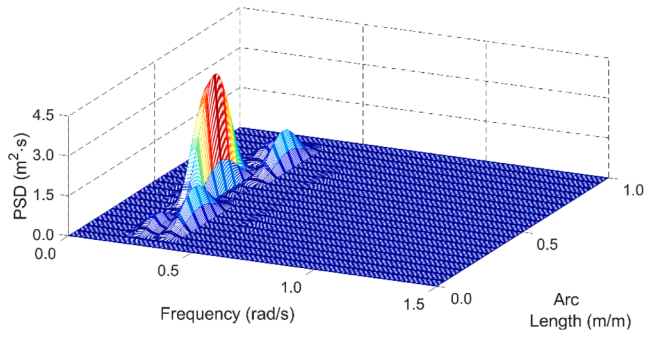
(b) Case 1 with TMD



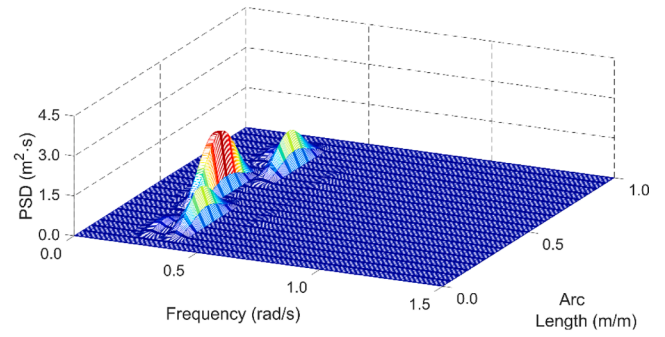
(c) Case 2 without TMD



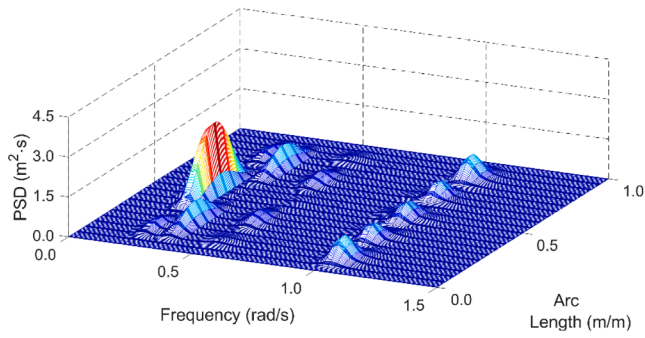
(d) Case 2 with TMD



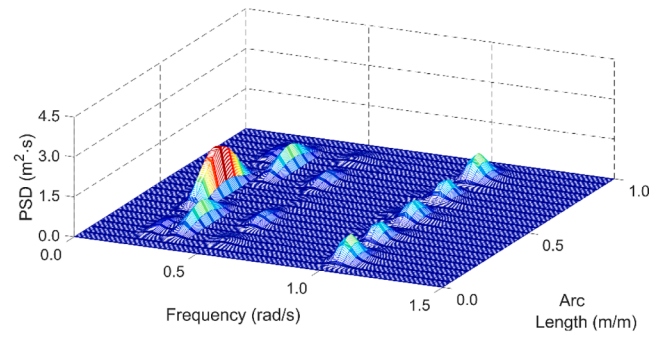
(e) Case 3 without TMD



(f) Case 3 with TMD



(g) Case 4 without TMD



(h) Case 4 with TMD

Fig. 11. Envelopes of PSDs of the girder's lateral motions with (b, d, f, and h) and without TMD (a, c, e, and g).

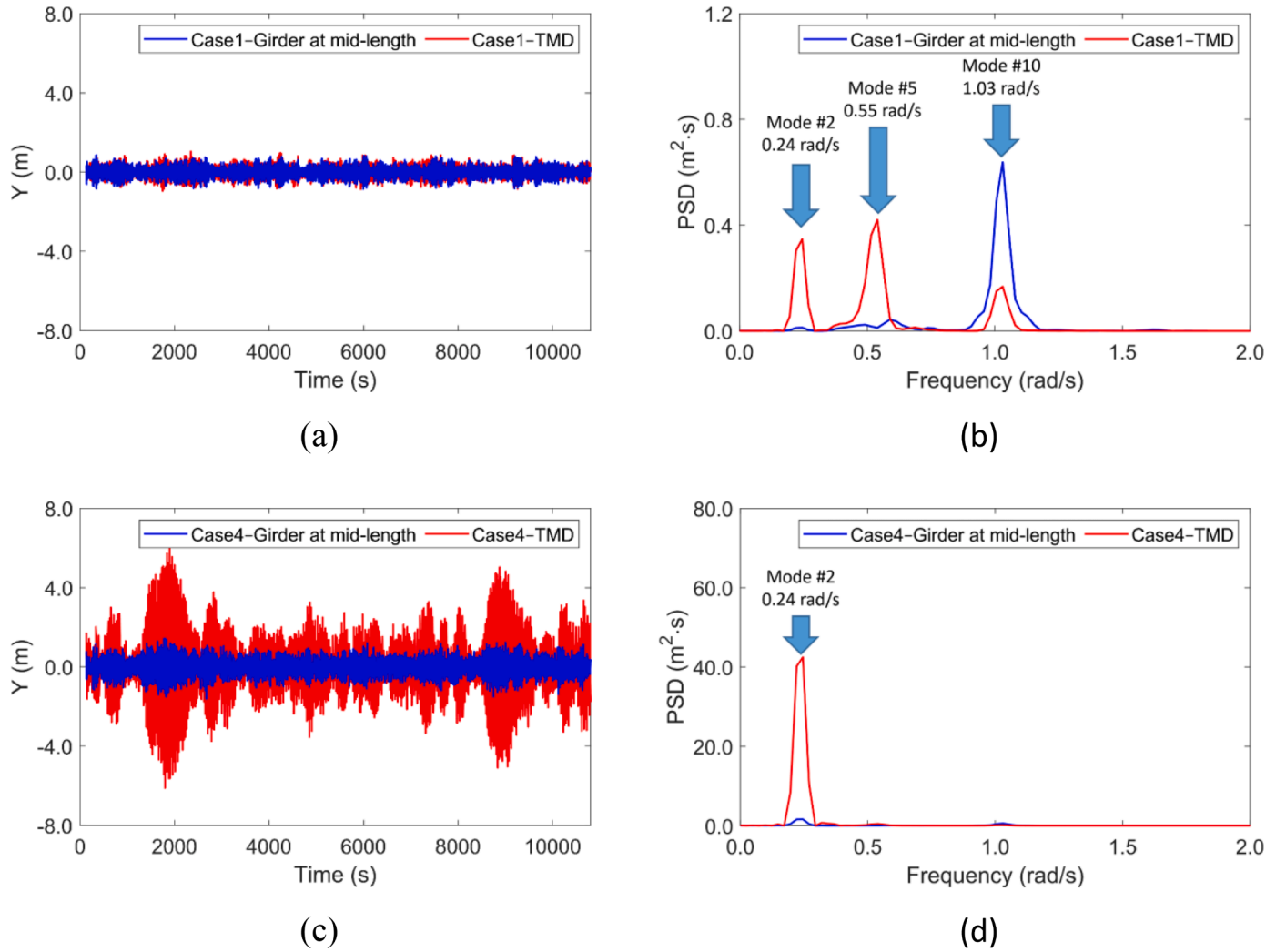


Fig. 12. Representative time histories (a and c) and PSDs (b and d) of lateral motion of the girder's mid-length and TMD.

uniformly distributed in $[0, 2\pi]$. Each regular wave has an equal amplitude, $\sqrt{2S(\omega_j)\Delta\omega_j}$, called the equal energy approach, which leads to varying frequency intervals $\Delta\omega$ depending on the spectrum shape. Signal repetition of a long time-history signal can be prevented by this method. The time history of wave elevation and original and recovered wave spectra are presented in Fig. 5.

Fig. 6 presents the optimization process. Population distributions shown in Fig. 6(a) show that the GA quickly finds global minima. In this case, the distribution narrows considerably to its optimal values after the 15th generation, as illustrated in Fig. 6(b). The obtained optimal spring and damping coefficients are 42.06 kN/m and 12.59 kNs/m, respectively. At the optimal spring coefficient, the natural frequency of TMD is 0.25 rad/s, which is very close to the second wet natural frequency of the floating bridge. As it was found in a previous study, in which the second wet natural frequency was the dominant mode (C. Jin et al., 2020), the optimal spring coefficient is close to this mode so that large resonant motion at 0.24 rad/s can be efficiently controlled. From now on, optimized TMD parameters are used to check its effectiveness.

Fig. 7 compares the envelope of the girder's lateral motion with and without TMD under white noise wave excitations. Maximum/minimum and standard deviation demonstrate the effectiveness of TMD in vibration control. Under a fixed-fixed boundary condition resulting in zero displacements at both ends, the largest responses occur at the mid-length. The maximum lateral motion decreases from 1.26 m to 0.82 m at the center, representing a percentage difference of 42.54%. Similarly,

the standard deviation of lateral motion is significantly reduced from 0.47 m to 0.28 m at the center, indicating a percentage difference of 50.72%. In addition, the second mode shape at a wet natural frequency of 0.24 rad/s, as shown in Fig. 2, closely resembles the standard deviation envelope, which means that the second mode plays an important role in the resonant motion of lateral motion, which is effectively controlled by TMD.

Fig. 8 shows the comparison between the lateral motion of the girder at mid-length and TMD. During instances of large resonant motion, TMD effectively controls the motion with a phase difference of approximately 90° , aligning with the typical TMD theory in which the phase angle of TMD is -90° compared with the main structure. Moreover, the lateral motion of TMD exceeds that of the girder's mid-length significantly, with the substantial inertia force of the TMD induced by large lateral motions governing the girder's lateral motion. The maximum TMD motion amplitude reaches around 4.2 m, surpassing practical limitations associated with the limited stroke of springs and dampers. Villoslada et al. (Villoslada and Santos, 2022) investigated vibration control effects with respect to TMD stroke. The results show that the longer the stroke, the better the vibration suppression. Their study even featured maximum strokes longer than the present one. In this regard, the near-maximum performance is achieved in the present study by allowing large strokes. Since this study focuses on the applicability of TMD with larger strokes, the stroke limit will further be considered as an additional constraint in detailed design.

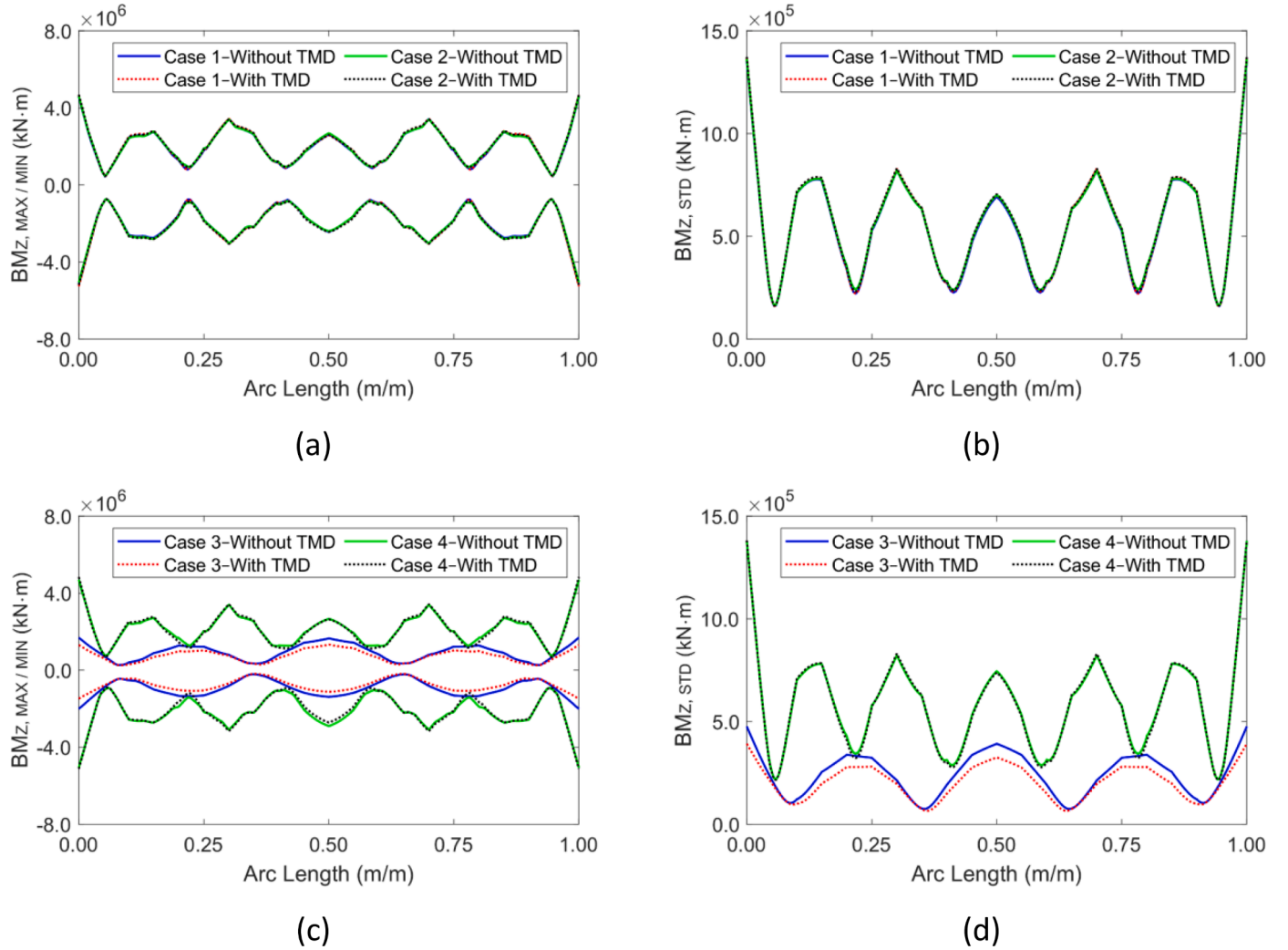


Fig. 13. Envelopes of maxima and minima (a and c) and standard deviations (b and d) of girder's vertical bending moments with and without TMD in Cases 1–4.

5. Effect of tuned mass damper under various combinations of environmental loads

TMD was further applied to the floating bridge under various combinations of environmental loads of wind, wave, and current, as defined in Cases 1–4 in Table 3. Environmental conditions in the Norwegian Sea around Fjord were employed. 100-year storm waves have a significant wave height and peak period of 3 m and 6 s while those of ocean swell waves are 0.4 m and 12 s. The Jonswap wave spectrum was used to generate time histories of wave elevations, by linearly superposing 100 regular waves for each wave condition. Enhancement parameters of the storm and swell waves were assumed to be 3.3 and 7, respectively. Again, similar to the white-noise wave spectrum, the equal energy approach was employed to avoid signal repetition for a long time-history signal. Maximum and minimum cut-off frequencies were 0.5 and 10 times peak frequency. The present study considers homogeneous waves to assess the maximum vibration control effect by TMD. Several studies have investigated the effect of inhomogeneous waves (Z. Cheng et al., 2018; Cui and Cheng, 2022; Dai et al., 2020), indicating slight changes in dynamic behaviors when considering inhomogeneous waves. This aspect will be a subject for future research. A one-year wind condition has a mean velocity of 22.9 m/s at a reference height of 10 m above the mean water level. The time history of wind velocity was generated by the API (The American Petroleum Institute) wind spectrum by superposing 1000 sinusoids. A constant current with a surface current velocity

of 0.7 m/s was also inputted. The direction of all environmental conditions was fixed to be 270°, which is perpendicular to the girder's longitudinal direction as defined in Fig. 1. A 3-hour simulation was conducted to better represent statistical values since second-order difference-frequency wave and wind loads induce slow varying lateral motions at the second wet natural frequency. Fig. 9 shows the time histories and spectra of wave elevation and wind velocity. The recovered spectra from FFT coincide with the original ones, demonstrating the correctness of produced time histories.

Fig. 10 shows the envelopes of maxima, minima, and standard deviations of the girder's lateral motion with and without TMD. The results of Cases 1–4 indicate that TMD is particularly effective in mitigating vibrations at the girder's mid-length, while there are negligible TMD effects around both ends, which coincides with the results under white-noise excitations. Fig. 10(a–b) shows that TMD is ineffective under first-order wave load, while it begins to operate as the second-order wave load is further accounted for. However, its functionality is not fully realized. On the other hand, as shown in Fig. 10(c–d), substantial decreases in lateral motion by TMD are observed when wind load is additionally considered. Table 4 provides the statistical values of lateral motion at the girder's mid-length. Percentage differences of maxima, minima, and standard deviation at the girder's mid-length in Case 3 are 31.2%, 25.3%, and 30.1% while those in Case 4 are 21.1%, 13.7%, and 12.4%, demonstrating the effectiveness of TMD in lateral vibration control.

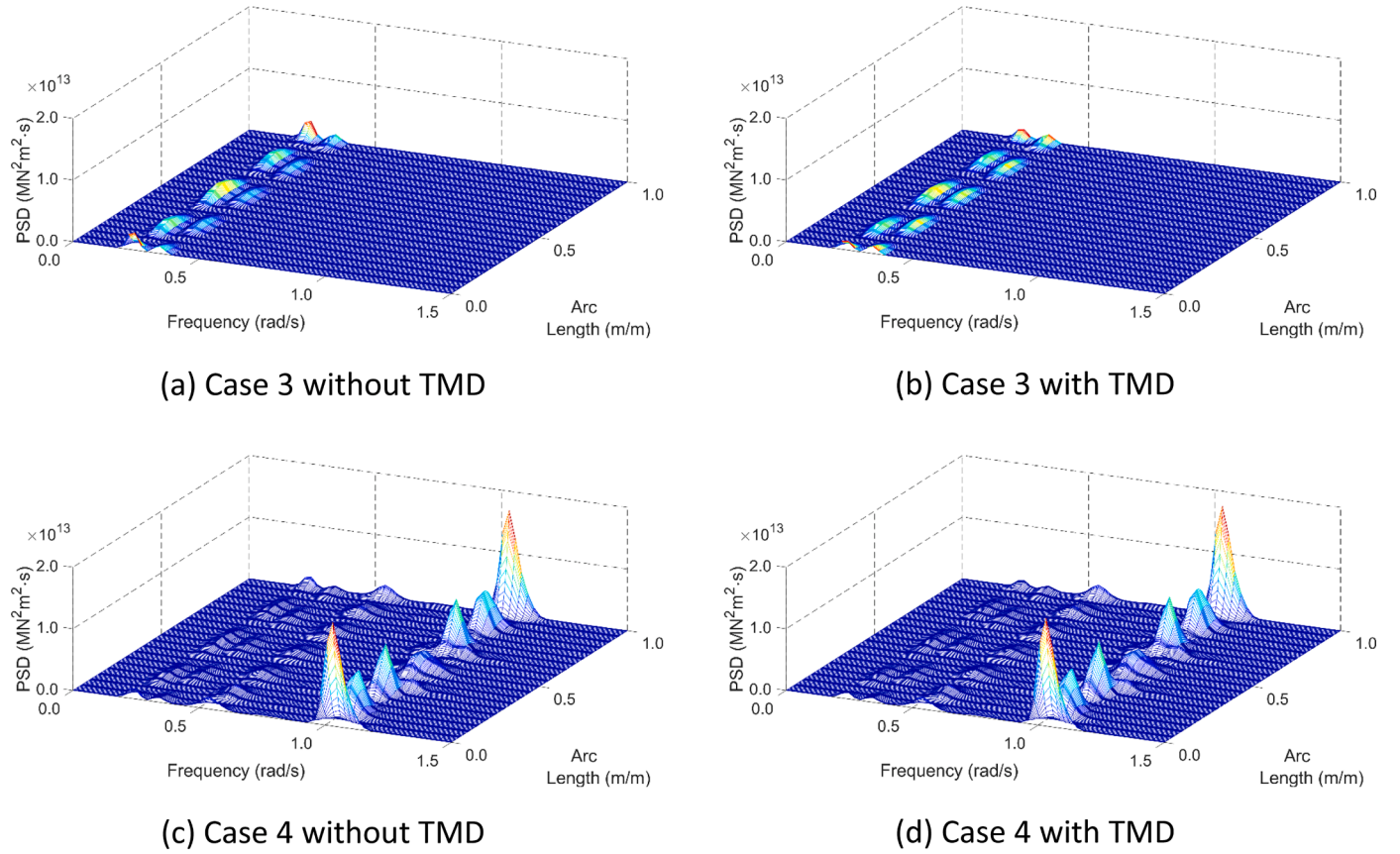


Fig. 14. Envelopes of PSDs of the girder's vertical bending moment with (b, d) and without TMD (a and c) in Cases 3–4.

To better check TMD's function in detail, time histories and spectra of the girder and TMD are further investigated. Fig. 11 shows the envelopes of power spectral densities (PSDs) of the girder's lateral motions with and without TMD. As shown in Fig. 11(a–b), without TMD, there are two dominant frequency peaks, one at swell waves around 0.52 rad/s and one at storm waves around 1.05 rad/s. The PSD shape due to the swell waves is similar to the fifth mode shape at 0.55 rad/s while that due to storm waves is similar to the tenth mode shape at 1.03 rad/s. It means that resonant motions at higher modes are observed under both swell and storm waves. The TMD effect is not observed in Case 1 since there is no motion energy at the second wet natural frequency of 0.24 rad/s under the first-order wave load. The second-order difference-frequency wave load induces lateral motion at 0.24 rad/s, which is controlled by TMD to a small degree, as Fig. 11(c) is compared with Fig. 11(d). Wind loads cause a strong resonance peak at 0.24 rad/s since wind velocity is dominant in the low-frequency regions, which is well controlled by TMD as demonstrated in Fig. 11(e–f). In addition, Fig. 11(g–h) shows that when we consider all loading components, TMD only plays an important role in decreasing resonant motion at the second wet natural frequency of 0.24 rad/s, while TMD is not functional in other frequency regions.

Fig. 12 shows representative time histories and corresponding PSDs of lateral motion of the girder's mid-length and TMD. In Case 1, the magnitudes of lateral motion of the girder's mid-length and TMD are similar, as shown in Fig. 12(a). Similar to Fig. 11(b), spectral plots shown in 12(b) indicate that the girder's lateral motion has two dominant peaks around the storm and swell wave frequencies of 1.05 rad/s and 0.52 rad/s, while very small resonant peak due to second wet natural frequency is observed around 0.24 rad/s. Spectral plots of TMD lateral motion show similar peaks. Even though TMD controls lateral motion around 0.24 rad/s, the girder's lateral motion is too small at that frequency, which results in limited motion control ability. However, in

Case 4, TMD's lateral motion is much larger than the girder's lateral motion. Spectral plots show that the girder's lateral motion has a high spectral peak at the second wet natural frequency, and to control this resonant motion, TMD has a very large peak at the second wet natural frequency.

The results presented in Figs. 10–12 indicate that the girder's lateral motion is well controlled when resonant motion at the second wet natural frequency of 0.24 rad/s appears, which is mainly induced by the wind load. Since motion or acceleration is highly related to passengers' safety and comfort, the TMD application will increase the safety of people when they pass through the floating bridge.

Finally, Figs. 13–14 show envelopes of the girder's bending moment with respect to the z-axis (referred to as vertical bending moment) with and without TMD. While TMD effectively controls lateral motions under wind loads, its impact on controlling bending moment is less significant. Since the vertical bending moment is associated with a curvature such that $BM_z = EI_z \kappa_z$ where E is Young's module, I_z is the second moment of area with respect to the z-axis, and κ_z is curvature with respect to the z-axis, lateral motion at higher modes can result in a higher magnitude of curvature and, consequently, a higher bending moment. Therefore, vertical bending moments are significant at the frequency around storm waves of 1.05 rad/s. As TMD primarily targets low-frequency motions around 0.24 rad/s, its effectiveness is not evident within the dominant frequency range of storm waves. Consequently, its efficacy in controlling vertical bending moment is limited. Results imply that structural safety related to fatigue of the system is not significantly improved by TMD when TMD is targeting lateral motion control. To both control lateral motion and vertical bending moment at multiple resonant frequencies, multiple TMDs can be connected in series at mid-length, or multiple TMDs can be separately located at different arc lengths. In addition, considering that the natural frequencies of floating bridges are lower than those of typical bridge structures, the TMD's natural frequency

needs to be low, which can lead to a heavier TMD mass. The next study will further investigate these critical factors.

6. Conclusions

This study proposes a passive TMD to control the lateral vibration of a curved floating bridge. The complex interaction among the girder, columns, pontoons, and TMD was analyzed successfully using a dynamic simulation model in the time domain. GA was utilized to optimize the spring and damping coefficients of TMD under white-noise wave excitations, and the fitness function was the summation of the standard deviation of the sway motion of 19 pontoons. The optimal results were achieved in a time-efficient manner across only a few generations of the population. Next, with optimized TMD coefficients, its functionality was further evaluated under a variety of loading combinations of wind, first-order wave, second-order difference-frequency wave, and current loads. The following key results are obtained:

- The optimal spring and damping coefficients obtained by GA were aimed at controlling vibration in the bridge's second wet mode in the lateral direction, known to be the dominant mode in the lateral motion of the curved floating bridge.
- The present TMD plays a significant role in targeting only one resonance peak, which is the second wet natural frequency.
- While TMD effectively controls wind-induced lateral motions at the second wet natural frequency, it shows limited effectiveness in mitigating other loading components, such as first- and second-order difference-frequency wave loads.
- The TMD does not effectively control the vertical bending moments, primarily because the curvature is governed by the higher mode around the wave frequency range, rather than the natural frequency of the floating bridge.

Future research will focus on exploring alternative TMD arrangements. Introducing multi-DOF TMDs in series at the bridge's mid-length or implementing multiple TMDs across the bridge girder could potentially offer control over multiple natural frequencies, thereby tackling the challenges of diverse low- and high-frequency excitations. As such, future studies aim to push the boundaries of our current understanding and seek out the most effective solutions for the complex issue of bridge vibration control.

CRedit authorship contribution statement

Chungkuk Jin: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Sewon Kim:** Writing – review & editing, Investigation. **Woo Chul Chung:** Validation, Supervision, Software, Methodology. **Byungho Kang:** Writing – review & editing, Visualization, Validation. **Seungjun Kim:** Data curation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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